

MIERCUREA CIUC, Romania

WMO: 151700

Lat: 46.3714N

Lon: 25.7725E

Elev: 2173

StdP: 13.58

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-2.8	-11.6	3.2	-8.2	-5.6	4.4	-1.7	19.2	33.2	17.2	31.7	1.3	330	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	85.3	65.2	81.9	64.6	78.9	63.4	68.0	79.5	66.3	78.0	64.8	76.0	6.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	97.5	72.5	62.4	91.3	70.8	60.8	86.3	68.8	33.6	79.6	32.2	78.0	31.0	76.5	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	14.3	12.2	-16.1	89.6	6.8	3.0	-21.0	91.8	-25.0	93.6	-28.8	95.2	-33.8	97.4		
(5)			-16.3	70.6	6.8	2.1	-21.2	72.1	-25.2	73.3	-29.0	74.5	-33.9	76.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	21.1	24.8	34.0	45.2	53.9	60.4	63.0	62.7	54.3	44.7	34.5	23.9		
	DBStd	16.79	11.08	10.95	8.73	6.11	5.83	5.01	4.68	4.90	5.97	6.83	8.16	10.13		
	HDD50	3785	896	704	498	164	31	1	0	0	27	188	466	810		
	HDD65	7871	1361	1124	962	594	346	150	93	102	322	628	914	1275		
	CDD50	1467	0	0	1	20	150	314	404	394	157	25	2	0		
	CDD65	78	0	0	0	0	0	13	32	31	2	0	0	0		
	CDH74	2069	0	0	0	13	85	350	692	757	166	6	0	0		
	CDH80	514	0	0	0	1	5	62	193	217	35	1	0	0		
(14) Wind	WSAvg	3.7	3.1	3.3	4.4	4.4	4.2	4.0	4.2	3.5	3.7	3.1	3.1	2.9		
(15) Precipitation	PrecAvg	22.3	1.1	1.0	1.1	1.7	2.8	3.6	3.3	2.6	1.6	1.3	1.1	1.1		
	PrecMax	30.1	3.1	3.3	3.3	2.9	6.0	8.5	6.7	5.2	4.9	3.9	3.3	2.4		
	PrecMin	16.4	0.0	0.0	0.0	0.7	1.2	0.7	0.5	1.1	0.0	0.2	0.1	0.1		
	PrecStd	3.3	0.7	0.7	0.7	0.7	1.1	1.6	1.4	1.2	1.1	0.9	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	52.6	64.7	75.5	80.1	85.9	89.5	90.2	84.6	72.9	61.7	49.3		
		MCWB	41.1	43.8	49.3	55.7	59.7	65.5	64.8	66.4	61.9	58.0	52.1	44.7		
	2%	DB	40.8	46.0	57.4	69.2	76.3	81.3	85.1	85.4	78.7	68.0	55.9	42.8		
		MCWB	37.7	40.7	45.9	53.2	58.9	64.7	65.3	65.8	61.2	55.6	49.4	40.3		
	5%	DB	37.6	42.1	52.3	63.9	72.7	78.2	81.4	81.7	74.3	63.0	51.2	39.2		
		MCWB	35.5	38.2	43.2	49.9	57.9	63.2	65.1	64.8	59.3	53.7	46.4	37.3		
	10%	DB	34.7	38.6	47.5	59.9	69.2	74.7	77.5	78.1	69.8	58.4	47.0	35.8		
		MCWB	33.1	35.8	41.0	48.6	56.6	62.0	63.9	63.8	57.9	51.5	43.6	34.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	45.0	50.3	56.8	62.9	69.2	69.9	70.5	65.1	59.7	53.0	45.1		
		MCDB	44.5	50.8	61.7	72.7	74.6	81.0	80.6	82.1	79.1	70.8	59.8	48.6		
	2%	WB	38.3	41.0	46.7	54.1	60.9	66.6	67.8	68.1	62.6	56.9	50.0	40.4		
		MCDB	40.3	45.3	55.7	66.7	72.0	77.5	78.6	80.5	75.0	65.4	55.3	42.6		
	5%	WB	35.6	38.5	44.0	51.7	59.2	64.7	66.2	66.2	60.6	54.6	46.9	37.5		
		MCDB	37.3	41.8	51.1	61.6	70.1	75.1	77.6	78.2	71.6	61.8	50.6	39.1		
	10%	WB	33.3	35.9	41.4	49.6	57.6	63.0	64.7	64.6	58.6	52.1	43.9	34.6		
		MCDB	34.6	38.4	46.7	57.9	67.6	72.5	75.7	76.0	68.4	57.9	46.8	35.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	18.7	20.5	24.0	25.1	24.9	26.2	27.7	26.3	24.6	17.2	13.2		
		MCDBR	14.5	20.6	30.5	34.7	33.5	32.4	34.5	35.6	36.9	33.9	25.7	15.4		
	5% WB	MCWBR	11.5	14.8	18.9	19.1	17.2	16.6	16.6	17.6	19.9	21.9	18.1	12.6		
		MCWBR	13.4	19.5	26.7	30.2	29.8	29.1	29.5	31.8	32.1	31.1	23.4	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.344	0.393	0.458	0.430	0.425	0.447	0.459	0.410	0.369	0.340	0.312		
		taud	2.276	2.194	2.136	2.008	2.164	2.221	2.166	2.124	2.222	2.325	2.340	2.315		
	Ebn at Noon		244	257	260	252	266	267	259	250	252	246	231	229		
		Edh at Noon	25	34	42	52	46	43	45	46	38	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		413	676	1033	1384	1726	1853	1852	1694	1210	816	456	320		
	RadStd		53	77	115	136	155	159	155	143	130	98	53	38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	N/A	N/A	+1.67	+0.88	N/A	-378	-524	+202	+34
(47) Regional (64 neighbors)		+1.19	N/A	N/A	+1.40	N/A	N/A	-263	-360	+140	+34

Nomenclature: See separate page